Application Note

AN102: AES70 Control Grouping

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Introduction

This application note describes AES70"s scheme for control grouping - using one control to affect multiple parameters, as in mastering, submastering, and so on.

The OcaGroup Class

The new class OcaGroup in AES70-2024 gives developers an easy to use way of defining control groups for master and submaster gain and EQ controls, mute groups, solo functions, and other interesting functions involving multiple control objects. It replaces an older class OcaGrouper , which was more difficult to use.

An OcaGroup control group is a list of objects ("members"). Changes to the group"s settings affect the settings of all its members. The exact relationship between group settings and member settings depends on a rule called the ***Aggregation Rule*** and will vary, depending on what kinds of objects are grouped.

- **Example 1 - Master mute.** Group members are OcaMute objects. Muting the group mutes all the individual objects, unmuting the group unmutes all the individual objects.
- **Example 2 - Virtual VCA master.** Group members are OcaGain objects. The group gain setting is added to the individual gain settings.

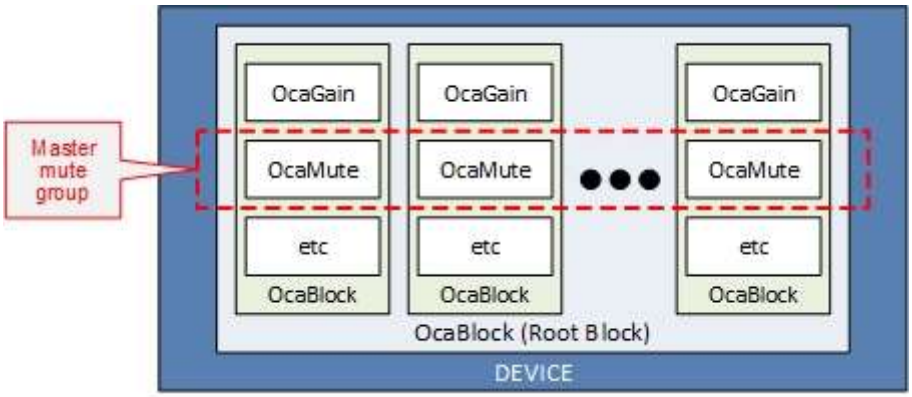
Control Groups and OcaBlocks

Many devices use OcaBlock objects to organize their control structures - see application notes AN100:Designing an AES70 Device for an overview of this technique, and AN101: OcaBlock Deep Dive for more details on OcaBlock .

It"s important to realize that OcaGroup control group membership is independent of a device"s block structure. The members of a given control group can belong to various blocks. If you"re familiar with conventional mixing console

architecture, you can think of AES70 control groups as functionally equivalent to VCA masters.

Here"s a multichannel mixer example showing these relationships.



Example AES70 control group

Definitions

The official description of AES70 control groups is in the AES70-1 standard, and the normative definition of the OcaGroup class is in the AES70 class definitions in AES70-2, Annex A.

References

AES70 standards

The following Audio Engineering Society standards define the AES70 Suite:

ID	Current Version	Name
AES70-1	2024	AES standard for audio applications of networks - Open Control Architecture - Part 1: Framework
AES70-2	2024	AES standard for audio applications of networks - Open Control Architecture - Part 2: Class structure
AES70-3	2024	AES standard for audio applications of networks - Open Control Architecture - Part 3: OCP.1 Binary protocol
AES70-4	2026	AES standard for audio applications of networks - Open Control Architecture - Part 4: OCP.2 JSON-based protocol
AES70-21	2026	AES standard for audio applications of networks - Open Control Architecture - Part 21: Using AES70 to manage AES67 and SMPTE ST 2110 30 media transport
AES70-22	2024	AES standard for audio applications of networks - Open Control Architecture - Part 22: Using AES70 to manage Milan™ media transport
AES70-23	2026	AES standard for audio applications of networks - Open Control Architecture - Part 23: Using AES70 to manage Dante® media transport

Copies of these standards are available from the AES Standards Store, aes2.org/publications/standards-store.

AES standards documents are free to AES members. You"re encouraged to join!

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